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Bachelor Thesis

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Faculty of **Informatik**

Implementation and Visualization of a Post-Quantum Cryptosystem Using the Example of the McEliece Cryptosystem

First- und Last name : **Thomas Schneider**

Registered on : **[TODO: Anmeldedatum]**

Submitted on : **[TODO: Abgabedatum]**

First Examiner : **[TODO: Erstprüfer/in]**

Second Examiner : **[TODO: Zweitprüfer/in]**

AFFIDAVIT

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Ingolstadt, _____

Thomas Schneider

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Thomas Schneider
Ingolstadt, Germany
[TODO: Monat Jahr]

ABSTRACT

Here goes a brief description of your entire thesis.

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Ingolstadt, _____
(Date)

(Signature)
Thomas Schneider

ACRONYMS

CTO CrypTool Online.

ECC Elliptic Curve Cryptography.

NIST National Institute of Standards and Technology.

PQC Post-Quantum Cryptography.

RSA Rivest-Shamir-Adleman.

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1 INTRODUCTION

1.1 Motivation

For decades, the security of widely deployed public-key cryptosystems such as RSA and Elliptic Curve Cryptography (ECC) has rested on the assumed intractability of integer factorization and the discrete logarithm problem for classical computers. [Shor \(1997\)](#) showed that both problems can be solved in polynomial time on a sufficiently large quantum computer, meaning that a cryptographically relevant quantum computer would render these widely used schemes insecure. Although such a machine does not yet exist, the risk that data encrypted today could be decrypted retroactively once quantum computers become available – commonly referred to as “store now, decrypt later” – has already prompted standardization bodies to act. In August 2024, the National Institute of Standards and Technology (NIST) published the first three finalized post-quantum cryptography (PQC) standards ([National Institute of Standards and Technology \(NIST\), 2024](#)), and the German Federal Office for Information Security (BSI) recommends that organizations begin migrating to quantum-safe algorithms without delay ([Bundesamt für Sicherheit in der Informationstechnik \(BSI\), 2021](#)). Post-quantum cryptography has thus moved from a theoretical research topic to an active engineering concern.

Among the algorithm families studied for this purpose, code-based cryptography is one of the oldest. The McEliece cryptosystem, introduced by [McEliece \(1978\)](#) almost five decades ago, has withstood cryptanalysis ever since and is still regarded as one of the most conservative choices for post-quantum encryption ([Bernstein et al., 2024](#)). Its most recent NIST submission, Classic McEliece, was not selected for standardization in the fourth round of the NIST PQC process – not because of a security weakness, but primarily because of its comparatively large public keys and to avoid diverging from a parallel ISO standardization effort ([Alagic et al., 2025](#)). This makes McEliece a notable case: a cryptosystem that is simultaneously among the most trusted and among the least understood by students, since its security relies on coding-theoretic concepts (linear codes, Goppa codes) that are typically not part of an undergraduate curriculum ([Overbeck and Sendrier, 2009](#)).

Interactive, browser-based learning tools have proven effective at making other areas of cryptography more approachable. CrypTool Online (CTO) already provides such visualizations for a range of classical and modern algorithms. However, no comparable visualization currently exists for the McEliece cryptosystem within CTO, leaving a gap between the system’s practical and historical importance and the resources available to teach it.

1.2 Research Question

Building on this motivation, this thesis addresses the following research question:

How can the McEliece cryptosystem be implemented and visualized within CrypTool Online to improve the understanding of code-based post-quantum cryptography for students?

1.3 Structure of the Thesis

The remainder of this thesis is structured as follows. section 2 briefly introduces the cryptographic fundamentals needed as a common frame of reference. section 3 covers the fundamentals of quantum computing and surveys the post-quantum cryptography landscape, positioning McEliece within it. section 4 presents the McEliece cryptosystem itself, from the underlying coding theory to key generation, encryption, and decryption. section 5 introduces CrypTool Online and its plugin architecture as the technical platform for this work, and reviews related educational tools. section 6 develops the educational and didactic concept underlying the planned visualization. section 7 derives the requirements for the plugin from the problem statement, objectives, and scope of this thesis. section 8 presents the design of the McEliece learning plugin, and section 9 describes its implementation within CTO. section 10 evaluates the resulting artifact, and section 11 concludes the thesis and outlines future work.

2 FUNDAMENTALS OF CRYPTOGRAPHY

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2.2 Security Concepts

2.3 Computational Hardness

2.4 Public-Key Cryptography

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10.5 Limitations of the Evaluation

11 CONCLUSION AND FUTURE WORK

11.1 Summary of Contributions

11.2 Answering the Research Question

11.3 Limitations

11.4 Future Work

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